Notes on Homological Algebra

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Last compiled at 2026-04-14

Contents

1. Notations 1
2. Derived functors 1
   2.1. Projective / Injective objects 1
   2.2. Projective / Injective resolutions 2
   2.3. Cohomology 2
   2.4. Comparison lemma 2
   2.5. Right dereived functor 3
   2.6. Injective objects are acyclic 4
   2.7. We love left-exact functors 4
   2.8. $(R^n F)_{n\geq 0}$ is a δ-functor 4
   2.9. F-acyclic objects 4
   2.10. Kunneth formula 5
   2.11. Universal ∂-functors 5
      2.11.1. Erasable functor 5
3. Derived category 5
   3.1. Mapping Cone and Cylinder 5
4. Sheaf cohomology 7
   4.1. Why you want flabby resolution 7
   4.2. Universal δ-functors are useful 8
   4.3. Cech cohomology $\cong$ sheaf cohomology 8
      4.3.1. $\check{H}$ is a δ-functor for presheaves 9
      4.3.2. $\check{H}$ is a δ-functor for sheaves 9
      4.3.3. $\check{H}$ is universal 9
   4.4. Singular cohomology $\cong$ sheaf cohomology 10
   4.5. Soft sheaves 10

Ref. Prof. Jean Gallier's book.

1. Notations

We agree these conventions if not stated otherwise.

$\mathcal{A}$ an abelian category;

2. Derived functors

2.1. Projective / Injective objects

We love these objects because the induced hom functors and tensor functors are exact. So it is nice when applying those induced functors to exact sequence.

But not all objects are projective or injective, so people are interested how far is an ordinary object from "good" objects (like projectives), and this is why we investigate resolutions.

> A common cause for need of resolutions occurs when in a category of nice objects certain universal constructions – such as quotients or intersections – fail to exist. Homotopical resolution may allow to embed the nice objects in categories of systems of nice objects (such as simplicial objects or chain complexes) inside which homotopical resolutions may be found (such as simplicial resolutions or homological resolutions, respectively).
>
> For example the quotient of a scheme or a smooth manifold by the action of an algebraic group or Lie group may fail to be a scheme or smooth manifold again, respectively, but the corresponding action groupoid (orbifold) serves as a resolution of the quotient in a suitable homotopical category of simplicial objects in schemes/manifolds (presenting the "quotient stack", see at higher geometry for more).
>
> — *Resolution* (nLab)

2.2. Projective / Injective resolutions

Definition 2.2.1 (Injective resolution)

Let $A \in \mathcal{A}$, an injective resolution of A is an exact sequence

$$0 \longrightarrow A \longrightarrow I^0 \longrightarrow I^1 \longrightarrow I^2 \longrightarrow \cdots$$

where I^k is injective for all $k \geq 0$.

We also denote this exact sequence by

$$0 \longrightarrow A \longrightarrow \boldsymbol{I}^A.$$

♣

The definition of a projective resolution is similar.

2.3. Cohomology

For an additive exact functor $F : \mathcal{C} \to \mathcal{D}$ between abelian categories and E an exact sequence in $\mathcal{C}$, the cohomology of $F(E)$ is trivial since F preserves exact sequence.

But if F is not exact, $F(E)$ is only a complex and the cohomology is not trivial in general. So in this point of view the cohomology groups $H^n(F(E))$ contain some information of exactness of F.

So we are interested in cohomology.

2.4. Comparison lemma

Morphisms between objects induce morphisms between injective resolutions.

Lemma 2.4.1 (Comparison lemma)

Let $g : A \to B$ be a morphism in $\mathcal{A}$, and $0 \to A \to \boldsymbol{I}^A$ and $0 \to B \to \boldsymbol{J}^B$ be injective resolutions.

Then we claim g induces a morphisms between resolutions:

$$\begin{tikzcd}
0 \arrow[r] \arrow[d, equal] & A \arrow[r] \arrow[d, "g"] & I^0 \arrow[r] \arrow[d, dashed, "g^0"] & I^1 \arrow[r] \arrow[d, dashed, "g^1"] & I^2 \arrow[r] \arrow[d, dashed, "g^2"] & \cdots \\
0 \arrow[r] & B \arrow[r] & J^0 \arrow[r] & J^1 \arrow[r] & J^2 \arrow[r] & \cdots
\end{tikzcd}$$

And any two lifts of g are chain homotopic.

♡

This lemma shows the uniqueness of injective resolution.

Theorem 2.4.2 (Injective resolution is unique)

Let $0 \to A \to \boldsymbol{I}^A$ and $0 \to A \to \boldsymbol{J}^A$ be two injective resolutions of $A \in \mathcal{A}$, then they are homotopic.

♡

Proof. By the previous lemma, $\mathrm{id}_A : A \to A$ induces two resolution morphisms: $g_{IJ} : \boldsymbol{I}^A \to \boldsymbol{J}^A$ and $g_{JI} : \boldsymbol{J}^A \to \boldsymbol{I}^A$.

Now $g_{IJ} \circ g_{JI} : \boldsymbol{J}^A \to \boldsymbol{J}^A$ is a lift of id_A, and $\mathrm{id}_J : \boldsymbol{J}^A \to \boldsymbol{J}^A$ is also a lift of id_A, so by the previous lemma, $g_{IJ} \circ g_{JI}$ is chain homotopic to id_J.

By the same argument $g_{JI} \circ g_{IJ} \simeq \mathrm{id}_I$. We are done. □

There are totally similar results for projective resolutions.

2.5. Right dereived functor

With the preparation in the previous section, we can now define the right dereived functors.

Definition 2.5.1 (Right dereived functor)

Let $F : \mathcal{C} \to \mathcal{D}$ be an additive functor between abelian categories and $\mathcal{C}$ admits enough injectives, then its n-th right dereived functor $R^n F : \mathcal{C} \to \mathcal{D}$ is defined as

$$C \mapsto H^n\left(F\boldsymbol{I}^C\right)$$

$$(g : C_1 \to C_2) \mapsto F\left(\left(g^i\right)_{i \geq 0}\right)$$

where $\boldsymbol{I}^C$ is one of the injective resolutions of C, and $\left(g^i\right)_{i \geq 0}$ is the induced map between injective resolutions.

♣

Remark

For any two injective resolutions $\boldsymbol{I}^C$ and $\boldsymbol{J}^C$, the previous theorem ensures they are homotopic and thus their cohomology groups are isomorphic.

For $g : C_1 \to C_2$ a morphism in $\mathcal{C}$, the previous lemma also ensures different lifts of g are chain homotopic so the induced maps between cohomology groups are the same.

These two observations ensures that $R^n F$ is well-defined for all $n \geq 0$.

2.6. Injective objects are acyclic

Theorem 2.6.1 (Injective objects are acyclic)

For $I \in \mathcal{A}$ injective and $F : \mathcal{A} \to \mathcal{B}$ additive, we have

$$R^n F(I) = 0$$

for all $n \geq 1$.

And $R^0 F(I) \cong F(I)$.

2.7. We love left-exact functors

Theorem 2.7.1 (F left-exact $\Longrightarrow R^0 F \cong F$)

As the title shows.

2.8. $(R^n F)_{n \geq 0}$ is a δ-functor

Theorem 2.8.1 ($(R^n F)_{n \geq 0}$ is a δ-functor)

This means for any exact sequence

$$0 \to A' \to A \to A'' \to 0,$$

there exists a bunch of morphisms $(\delta_n)_{n \geq 0}$ resulting a long exact sequence

$$\begin{tikzcd}
0 \arrow[r] & R^0F(A') \arrow[r] & R^0F(A) \arrow[r] & R^0F(A'') \arrow[dll, "\delta_0"'] \\
& R^1F(A') \arrow[r] & R^1F(A) \arrow[r] & R^1F(A'') \arrow[dll, "\delta_1"'] \\
& R^2F(A) \arrow[r] & R^2F(A) \arrow[r] & R^1F(A'') \arrow[dll, "\delta_2"'] \\
& \dots & &
\end{tikzcd}$$

Remark

Note $R^0 FX \cong FX$.

2.9. F-acyclic objects

Sometimes it is difficult to compute the cohomology groups of an injective resolution, but fortunately people have found another kind of objects may help us.

Definition 2.9.1 (F-acyclic objects)

Let $F : \mathcal{A} \to \mathcal{B}$ be an additive functor, an object $A \in \mathcal{A}$ is called F-acyclic if

$$R^n F(A) = 0$$

for all $n \geq 1$.

♣

Remark

1. Injective objects are F-acyclic for all functors $F : \mathcal{A} \to \mathcal{B}$;
2. We only require the $R^n FA$ to be trivial when $n \geq 1$, because $R^0 FA \cong A$ is determined.

This is the most important result of F-acyclic objects:

Theorem 2.9.2 (Compute $R^n F$ by F-acyclic resolutions)

♡

2.10. Kunneth formula

Kunneth theorem gives a exact sequence with which we can compute $H^n(A^\bullet \otimes B^\bullet)$ or the corresponding homology version.

An application:

$$H_p(T^n) = \mathbb{Z}^{\binom{n}{p}},$$

where $T^n := (S^1)^n$ is the n-tori.

2.11. Universal ∂-functors

2.11.1. Erasable functor

Ref. [Tohoku].

When F is erasable, it with its satellite functors form a universal ∂-functor.

3. Derived category

The derived category $D(\mathcal{A})$ of an abelian category $\mathcal{A}$ is the result of the chain complex category $\mathrm{Ch}(\mathcal{A})$ modding out quasi-isomorphisms.

3.1. Mapping Cone and Cylinder

Mapping cone and cylinder are two mysterious objects that play as the canonical gredient in a *distinguished triangle.*

Distinguished triangles play a role as exact sequence in abelian categories as in derived categories, since the latter, though additive, are not abelian and thus we can not talk about "exactness".

Definition 3.1.1 (Mapping cone)

For a map $f : K^\bullet \to L^\bullet$ between cochain complexes, its mapping cone is also a cochain complex

$$\mathrm{Cone}(f) := K^\bullet[1] \oplus L^\bullet,$$

and the differential map (aka coboundary map) $d : K^{p+1} \oplus L^p \to K^{p+2} \oplus L^{p+1}$ is given by

$$d(k^{p+1}, l^p) := ()$$

TODO: complete.

♣

Definition 3.1.2 (Mapping cylinder)

TODO

♣

The mapping cone and the mapping cylinder are related by the following lemma:

Lemma 3.1.3 (Cones and cylinders give "canonical" distinguished triangles)

For any map $f : K^\bullet \to L^\bullet$, there exists the following commutative diagram with exact rows:

$$\begin{tikzcd}
 & 0 \arrow[r] & L^\bullet \arrow[r, hook, "\text{inclusion}"] \arrow[d, "\alpha"] & \mathrm{Cone}(f) \arrow[r, two heads, "\delta_f"] \arrow[d, equal] & K[1]^\bullet \arrow[r] & 0 \\
0 \arrow[r] & K^\bullet \arrow[r, hook, "\text{inclusion}"] \arrow[d, equal] & \mathrm{Cyl}(f) \arrow[r, two heads, "\text{projection}"] \arrow[d, "\beta"] & \mathrm{Cone}(f) \arrow[r] & 0 & \\
 & K^\bullet \arrow[r, "f"] & L^\bullet & & &
\end{tikzcd}$$

where

- δ_f is the obvious projection (the reason to this strange name will be clear later);
- α is the obvious inclusion;
- $\beta(k^p, k^{p+1}, l^p) := f(k^p) + l^p$.

And further we have

- $\beta\alpha = \mathrm{id}_L$;
- $\alpha\beta$ homotopic to $\mathrm{id}_{\mathrm{Cyl}(f)}$.

♡

Proof. By direct computation, see also [Manin, III.3.3]. □

So what IS triangles?

Definition 3.1.4 (Triangle)

1. A triangle of complex is a diagram (may not be exact) like:

$$K^\bullet \longrightarrow L^\bullet \longrightarrow M^\bullet \longrightarrow K[1]^\bullet$$

2. A morphism between triangles is a triple of morphisms (f, g, h) such that the following diagram commutes:

$$\begin{tikzcd}
K_1^\bullet \arrow[r] \arrow[d, "f"] & L_1^\bullet \arrow[r] \arrow[d, "g"] & L_1^\bullet \arrow[r] \arrow[d, "h"] & K_1[1]^\bullet \arrow[d, "f[1]"] \\
K_2^\bullet \arrow[r] & L_2^\bullet \arrow[r] & M_2^\bullet \arrow[r] & K_2[1]^\bullet
\end{tikzcd}$$

   Two triangles are said to be isomorphic if f, g, h are all isomorphisms.

3. A triangle is said to be **distinguished** if it is isomorphic to a canonical triangle given by a map $f : K^\bullet \to L^\bullet$:

$$K^\bullet \xhookrightarrow{\text{inclusion}} \mathrm{Cyl}(f) \xrightarrow{\text{projection}} \mathrm{Cone}(f) \xrightarrow{\delta_f} K[1]^\bullet$$

♣

Now we can state the main theorem of cones and cylinders and triangles.

Theorem 3.1.5 (Shot exact sequence can be completed to a distinguished triangle)

An exact sequence

$$0 \hookrightarrow K \to L \twoheadrightarrow M \to 0$$

of complexes is quasi-isomorphic to

$$0 \to K \hookrightarrow \mathrm{Cyl}(f) \twoheadrightarrow \mathrm{Cone}(f) \to 0.$$

♡

Proof. The quasi-isomorphim is given by

$$\begin{tikzcd}
0 \arrow[r] & K \arrow[r, hook, "f"] \arrow[d, equal] & L \arrow[r, two heads, "g"] & M \arrow[r] & 0 \\
0 \arrow[r] & K \arrow[r, hook] & \mathrm{Cyl}(f) \arrow[r, two heads] \arrow[u, "\text{projection}"] & \mathrm{Cone}(f) \arrow[r] \arrow[u, "\gamma"'] & 0
\end{tikzcd}$$

where $\gamma(k^{p+1}, l^p) := g(l^p)$.

The projection is a quasi-isomorphim is already proved in Lemma 3.1.3.

As for γ, see [Manin, III.3.5]. □

4. Sheaf cohomology

4.1. Why you want flabby resolution

Theorem 4.1.1 (All injective sheaves are flabby)

All injective sheaves are flabby.

♡

Proof. Let $I \in \mathbf{Sh}_{\mathcal{A}}(X)$ be an injective sheaf, then consider its canonical flabby embedding

$$I \hookrightarrow \mathrm{Gode}(I),$$

since I is injective, we immediately obtain a retraction of this embedding and thus I is a direct summand of $\mathrm{Gode}(I)$.

It is "well-known" that every summand of a flabby sheaf is also flabby See [Jean, Prop. 13.6]. We are done. □

Theorem 4.1.2 (Flabby sheaves are acyclic)

Flabby sheaves are Γ-acyclic.

Proof. Flabbiness implies long exact sequence. □

Note $\mathbf{Sh}_{\mathcal{A}}(X)$ is an abelian category with enough injectives, and thus we can do (co)homology as in any other abelian categories.

With these two theorems in hand we can compute the cohomology groups of a sheaf using flabby resolutions instead of coomon injective resolutions.

Theorem 4.1.3 (Compute sheaf cohomology using flabby resolution)

Let $F \in \mathbf{Sh}_{\mathcal{A}}(X)$ and $R^i\Gamma(F)$ be the right derived functor applying on F, then

$$R^i\Gamma(F) \cong H^i(X, F)$$

where $H^i(X, F)$ is the cohomology groups of the complex defined by applying Γ to the canonical flabby resolution of F.

Proof. Because the derived functor R^iT can be computed by T-acyclic resolutions, see [Jean, Prop. 11.34]. □

4.2. Universal δ-functors are useful

Universal δ-functors are especially useful because of its uniqueness, which, in this section, will be used to prove the equivalence chain between various cohomology in algebraic topology, i.e. de Rham cohomology $H_{\mathrm{dR}}(X, -)$, singular cohomology $H_{\mathrm{Sing}}(X, -)$, Cech cohomology $\breve{H}(X, -)$ and sheaf cohomology $H(X, -)$.

The main idea is that, since the sheaf cohomology $\left(H^i(X, -)\right) : \mathbf{Sh}_{\mathcal{A}}(X) \to \mathcal{A}$ is an universal δ-functor, if we can prove that, for example, $\left(\breve{H}^i(X, -)\right)$ is also an universal δ-functor and $H^0(X, F) \cong \breve{H}^0(X, F)$ for any sheaf F, then by the uniqueness we have this remakable result:

$$H^i(X, -) \cong \breve{H}^i(X, -).$$

The same argument also applies to other cohomology theory.

4.3. Cech cohomology $\cong$ sheaf cohomology

By the uniqueness of universal δ-functor, it suffices to prove:

1. Cech cohomology $\breve{H} := \left(\breve{H}^i(X, -)\right)_{i \geq 0} : \mathbf{Sh}_{\mathcal{A}}(X) \to \mathcal{A}$ is an universal δ-functor.
   - $\breve{H}$ is a δ-functor for **presheaves**;
   - $\breve{H}$ is a δ-functor for **sheaves**, provided X is paracompact;
   - $\breve{H}$ is universal;
2. $\breve{H}^0(X, F) = H^0(X, F)$ for all sheaves F.

The second part is trivial by definition of Cech cohomology. The first part is rather involved.

4.3.1. $\check{H}$ is a δ-functor for presheaves

For a **presheaf** F, let $C^p(\mathcal{U}, F)$ be the set of p-cochains related to an open cover $\mathcal{U}$ of X. Then $C^p(\mathcal{U}, -) : \mathbf{PSh}_{\mathcal{A}}(X) \to \mathcal{A}$.

Proposition 4.3.1 ($C^p(\mathcal{U}, F)$ is exact for all $p \geq 0$)

♠

Proof. Unfold the definition of $C^p(\mathcal{U}, F)$ and this is trivial because exact sequence is stable under product. □

Collecting C^p we obtain a long exact sequence. And thus we reach:

Theorem 4.3.2

$\left(\check{H}^p(X, F)\right)_{p \geq 0}$ is a δ-functor for presheaves.

♡

Proof. Let $0 \to F' \to F \to F'' \to 0$ be an exact sequence of **presheaves**, then after applying $C^p(\mathcal{U}, -)$ we obtain a short exact sequence of **complex**:

$$0 \longrightarrow C^\bullet(\mathcal{U}, F') \longrightarrow C^\bullet(\mathcal{U}, F) \longrightarrow C^\bullet(\mathcal{U}, F'') \longrightarrow 0$$

Since exact sequence is stable under limit so

$$0 \longrightarrow C^\bullet(X, F') \longrightarrow C^\bullet(X, F) \longrightarrow C^\bullet(X, F'') \longrightarrow 0$$

is also exact.

By general technique this sequence of complex yields a long exact sequence of Cech cohomology groups. □

4.3.2. $\check{H}$ is a δ-functor for sheaves

The main idea is to prove that

$$\check{H}(X, F) \cong \check{H}(X, \mathbf{Sh}(F)),$$

where $\mathbf{Sh}(F)$ is the sheafification of presheaf F. And fortunately this is true when X is paracompact!

For the proof see [Jean, Prop 13.16].

4.3.3. $\check{H}$ is universal

To prove $\check{H} : \mathbf{Sh}_{\mathcal{A}}(X) \to \mathcal{A}$ is an universal δ-functor, we first recall the Grothendieck lemma:

Lemma 4.3.3 (Erasable δ-functors are universal)

Let $T = (T^n)_{n \geq 0} : \mathcal{C} \to \mathcal{D}$ be a δ-functor, if every object $A \in \mathcal{C}$ there exists a T-erasable monic $v : A \hookrightarrow M_A$, i.e. $T^n(v) = 0$ for all $n \geq 1$, then T is universal.

♡

Theorem 4.3.4 ($\check{H}(\mathcal{U}, -)$ is erasable at flabby embeddings)

For every space X, every open cover $\mathcal{U}$ of X, if the sheaf F is flabby then

$$\check{H}^i(\mathcal{U}, F) = 0, \forall i \geq 1.$$

♡

Proof. [Jean, Prop 13.12] □

Since $\breve{H}(X, -)$ is the colimit of $\breve{H}^i(\mathcal{U}, F)$, we have $\breve{H}(X, F) = 0$ for flabby F.

Note that $\breve{H}^0(X, F) \cong FX \cong H^0(X, F)$, so since they are both universal, we obtain the isomorphisms between them.

4.4. Singular cohomology ≅ sheaf cohomology

In this section we prove another meaningful isomorphism between different cohomology, but using a different method instead of proving H_{sing} is a universal δ-functor.

We first observe that $S^p(-, G) \in \mathbf{PSh}_{\mathbf{Grp}}(X)$ is only a **presheaf**. So in the following we need to consider the sheafification $\mathcal{S}^p(-, G)$ of it.

We will then prove that $(\mathcal{S}^p(-, G))_{p \geq 0}$ forms a Γ-acyclic resolution of the constant sheaf $\tilde{G}$ under some mild assumptions (when X is paracompact, $\mathcal{S}$ is soft and thus acyclic, but it is not flabby), and thus there is an isomorphism

$$\frac{\ker(\mathcal{S}^p \to \mathcal{S}^{p+1})}{\operatorname{im}(\mathcal{S}^{p-1} \to \mathcal{S}^p)} \cong H^p\big(X, \tilde{G}\big).$$

And the final step is to prove

$$H^p_{\text{Sing}}(X, G) \cong \frac{\ker(\mathcal{S}^p \to \mathcal{S}^{p+1})}{\operatorname{im}(\mathcal{S}^{p-1} \to \mathcal{S}^p)}.$$

We prove $\mathcal{S}$ is acyclic by proving it is soft.

4.5. Soft sheaves

A sheaf $F \in \mathbf{Sh}(X)$ is flabby if for every **open** U, the restriction map $FX \to FU$ is surjective.

Replace the open set in the definition of flabby sheaves by closed set, we obtain **soft** sheaves.